**7 (a)** The horizontal intercept of the graph shows the threshold wavelength above which no photoemission of electrons will take place. [A1]
Accept threshold frequency if a reference is made to $v = f\lambda$
Also accept: Maximum wavelength above which no emission of electrons will take place.

**(b) (i)** The minimum energy required can be found when the photon with the threshold wavelength causes photoemission – this is the x-axis intercept of the graph.

$$\text{Photon energy at threshold wavelength} = \frac{hc}{\lambda}$$

$$= \frac{6.63\times10^{-34}\times3\times10^{8}}{4.2\times10^{-7}} \quad \text{[C1]}$$

$$= 4.74\times10^{-19}\ \text{J} = 4.7\times10^{-19}\ \text{J} \quad \text{[A1]}$$

**(ii)** The kinetic energy of the emitted electron will be:

$$KE = \frac{hc}{\lambda} - \phi$$

$$= \frac{6.63\times10^{-34}\times3\times10^{8}}{2.0\times10^{-7}} - 4.74\times10^{-19} \quad \text{[C1]}$$

$$= 5.21\times10^{-19}\ \text{J} \quad \text{[A1]}$$

**(c)** Light is made of discrete quanta of electromagnetic radiation / packets of energy called photons. [B1]

One electron can only absorb the energy of one photon at a time. [B1]

Photons with wavelengths longer than the threshold wavelength / frequency lower than the threshold frequency do not have sufficient energy [B1]
for electrons to overcome the surface energy barrier to escape the metal.